

Supplemental Material:

Directed Distance Fields for Constant-Time Ray Queries on Gaussian Splatting

S1. PER-OBJECT E4 GI FIDELITY

This supplement complements Table IV of the main paper, which reports only aggregate means over our 142-object E4 benchmark. The table below lists, for every individual object, the shadow PSNR, ambient-occlusion (AO) PSNR, and AO mean absolute error (MAE) of the DDF-traced GI image against an embree mesh ray-cast reference (same primary trace, same lighting; only the secondary-ray oracle differs). Objects are grouped by subset (GSO first, then ShapeNet) and sorted alphabetically within each group.

Recap of aggregate means (from main paper Table IV): all 142 objects: shadow PSNR 30.34 dB, AO PSNR 21.27 dB, AO MAE 0.216; GSO (29): 30.81/21.29 dB, MAE 0.175; ShapeNet (113): 30.22/21.26 dB, MAE 0.227.

object	sh. PSNR (dB)	AO PSNR (dB)	AO MAE	object	sh. PSNR (dB)	AO PSNR (dB)	AO MAE
<i>GSO (29 objects)</i>				chair_19cbb7fd	18.91	12.77	0.443
airplane	20.61	13.20	0.368	chair_47eff1e5	20.98	14.32	0.385
allo	32.26	26.63	0.133	chair_54fc8ae4	19.62	13.52	0.451
bagel	27.80	14.88	0.322	chair_5f1b4529	28.46	23.38	0.209
blueMug	27.40	19.46	0.228	chair_b8666dca	18.97	14.20	0.379
boatShoe	35.91	20.97	0.192	clock_81be6024	27.59	11.37	0.413
bowl	27.64	18.76	0.207	clock_de0a333f	23.61	23.51	0.133
bull	27.54	21.75	0.153	dishwasher_21ee4122	39.38	25.84	0.044
bundtPan	25.94	13.30	0.342	dishwasher_e7323270	37.30	20.40	0.102
bus	22.90	15.49	0.227	display_a0383e13	30.00	23.48	0.117
car	38.11	24.08	0.161	display_b5f473b7	36.44	28.02	0.102
chocoBox	37.77	16.33	0.168	earphone_4de2413a	36.15	24.46	0.126
clock	35.28	28.56	0.043	faucet_b7758e9c	33.62	22.61	0.225
eagle	20.32	15.03	0.344	faucet_c4c8a8f2	14.76	10.14	0.625
hammer	49.21	24.75	0.190	file_c227d03f	35.23	18.80	0.254
horse	35.66	25.33	0.130	file_d879a859	22.02	18.99	0.272
lion	31.57	23.46	0.129	guitar_1edaab17	31.58	27.38	0.109
mug	24.05	15.44	0.253	guitar_2d31ac29	43.52	33.43	0.082
orca	30.69	23.60	0.133	helmet_3a12b9c4	33.60	28.56	0.020
panda	33.57	21.92	0.080	helmet_6f41517d	35.35	27.38	0.034
rhino	29.82	22.71	0.141	jar_bcacdde8	42.77	32.68	0.023
sausage	36.71	30.38	0.035	jar_ca00e439	21.23	12.51	0.298
shoe	31.85	21.65	0.177	knife_208610aa	41.95	45.30	0.012
spino	33.01	27.23	0.126	knife_6813197a	40.92	30.37	0.231
teapot	34.00	17.76	0.183	lamp_4078ba4c	30.60	25.93	0.140
teddy	26.69	25.80	0.087	lamp_5bc478e9	53.91	38.19	0.022
thomas	27.76	21.57	0.116	laptop_19117e74	18.75	19.97	0.171
torch	39.18	30.46	0.046	laptop_7f75b94b	14.96	14.32	0.291
triceratop	25.67	20.12	0.126	loudspeaker_11511510	21.11	14.45	0.319
turtle	24.69	16.69	0.227	loudspeaker_82b9111b	34.16	13.18	0.437
<i>ShapeNet (113 objects)</i>				mailbox_81f30bae	28.48	20.32	0.159
airplane_3209a020	27.81	22.04	0.148	mailbox_bfc242ae	29.33	17.03	0.253
airplane_54885d5b	30.89	21.51	0.295	microphone_1d4ac8a8	46.40	37.83	0.029
airplane_5ac00867	30.03	25.99	0.155	microphone_ae85334c	29.83	23.29	0.167
airplane_752a0bb6	25.42	21.13	0.260	microwave_31548a2f	19.87	10.85	0.455
airplane_94c4ade3	45.56	29.17	0.094	microwave_71221c38	14.12	10.56	0.430
airplane_d605a53c	29.60	25.30	0.170	motorcycle_9d3b07f4	27.88	20.43	0.297
ashcan_5a947a76	26.68	15.71	0.281	motorcycle_e59b0be8	31.48	24.39	0.216

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object	sh. PSNR (dB)	AO PSNR (dB)	AO MAE	object	sh. PSNR (dB)	AO PSNR (dB)	AO MAE
ashcan_8b071aca	29.52	23.99	0.082	mug_cc5b14ef	22.92	14.98	0.250
bag_506a05b6	32.68	12.73	0.366	mug_eecb13f6	38.06	23.27	0.060
bag_5afe3102	39.72	27.67	0.041	piano_89814e65	17.35	6.92	0.734
basket_34fd44c4	38.45	22.82	0.068	piano_a9292522	24.75	15.35	0.314
basket_dc4a523b	33.59	18.96	0.154	pillow_4ae37fde	52.91	26.21	0.050
bathub_1fb49b04	28.47	3.89	0.863	pillow_953d16ed	33.16	28.47	0.034
bathub_33718430	29.58	16.00	0.349	pistol_d3baf46b	31.72	25.12	0.137
bed_218a4d46	27.55	9.47	0.560	pistol_db298a7c	29.83	26.54	0.105
bed_75e308dc	20.78	16.84	0.469	pot_ea873ee1	24.74	17.42	0.164
bench_3d2ee152	21.53	16.05	0.284	pot_f1c17606	23.74	17.36	0.165
bench_dbacfce8	23.34	21.91	0.203	printer_4e2784b9	37.52	26.83	0.068
birdhouse_9c038a6c	30.88	27.97	0.139	printer_c73c7f57	22.14	16.30	0.193
birdhouse_be167ea4	32.67	19.56	0.194	rifle_74a38b74	28.66	23.35	0.194
bookshelf_1e46876d	27.31	19.07	0.293	rifle_efe36ad6	36.04	30.88	0.065
bookshelf_a9c2bcc2	30.82	19.64	0.256	rocket_60e62b41	50.99	37.85	0.053
bottle_59d7b4e7	51.00	42.76	0.009	rocket_b094ce52	33.92	27.77	0.143
bottle_9a70dd11	29.03	26.61	0.065	skateboard_36aaae33	40.13	27.87	0.096
bowl_4b32d2c6	20.28	12.44	0.360	skateboard_6e76a333	48.05	30.13	0.077
bowl_d5b3fb99	20.22	9.52	0.527	sofa_145bd097	20.48	13.22	0.397
bus_9178a769	32.39	18.42	0.339	sofa_bf816001	29.23	19.04	0.242
bus_9525a66b	31.43	22.01	0.148	stove_966ea3af	21.55	14.55	0.294
cabinet_2b3f95e3	25.69	10.59	0.442	stove_fdad15a5	23.50	16.84	0.161
cabinet_85ddb69	23.61	13.35	0.325	table_6086c619	31.37	13.21	0.329
camera_12986340	25.99	20.70	0.153	table_d2d5b8f9	20.45	14.46	0.333
camera_2c0b4e31	25.44	18.85	0.244	telephone_a262dd4b	42.25	35.54	0.024
can_8cf26f69	28.74	10.62	0.416	telephone_a523f6e5	33.05	14.39	0.577
can_d801d5b0	41.63	37.64	0.010	tower_35824a59	34.43	35.13	0.043
cap_4bd0b6df	23.48	22.97	0.194	tower_44e8212f	27.79	17.21	0.270
cap_5eb9ab53	21.19	15.60	0.271	train_aaa3e125	29.49	24.82	0.112
car_1328a95d	28.96	24.53	0.094	train_b9818859	56.20	35.86	0.038
car_517dd2b0	30.80	20.19	0.179	vessel_31ddf87b	24.66	19.81	0.323
car_9ee32f51	34.67	24.16	0.113	vessel_a7b07a2a	22.54	17.06	0.369
car_adc6f0f3	27.79	22.04	0.168	washer_d1aa6a6c	27.98	13.85	0.455
car_e9bdc6c0	28.10	16.56	0.293	washer_d78b4c3d	25.32	13.89	0.259

The aggregate means above match Table IV of the main paper. Higher PSNR and lower AO MAE indicate closer agreement with the embree mesh ray-cast reference; see Section 6 (E4) of the main paper for the benchmark protocol.